MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : ESC 301/ESC301/PC-ROB301 Analog and Digital Electronics

UPID : 003442

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) Which type of power amplifier is biased for operation at digital signal operation?
a) Class A b) Class B or AB c) Class C d) Class D

(II) The number of 2 i/p XOR gates required for the conversation of 11011 to its equivalent Gray Code are
(a) 2 (b) 4 (c) 3 (d) 5

(III) The invalid state for S-R latch occurs when S and R are _____.
a) both high
b) both low
c) S = high, R = low
d) S = low, R = high

(IV) Which of the following represents range of frequency measured by ADC?
a) Bandwidth
b) Threshold frequency
c) Peak frequency
d) None of the mentioned

(V) Which of these is a disadvantage of the Wien Bridge oscillator?
a) It cannot be used for high resistance values.
b) There is no automatic gain control.
c) Distortion observed in output is high.
d) It cannot fabricate a pure tune.

(VI) How many Half adders required to add two 8 bit numbers?
(a) 8 (b) 15 (c) 10 (d) 2

(VII) What is the difference between a shift-right register and a shift-left register?
a) There is no difference
b) The direction of the shift
c) Propagation delay
d) The clock input

(VIII) Which of the following method is employed for ADC? (1)
a) Ladder network
b) Successive approximation type
c) PWM type
d) None of the mentioned

(IX) The voltage gain of the amplifier is 8 and the current gain is 7. The power gain of the amplifier is.
a) 56 d
b) 17.481 db
c) 34.963 db
d) 1 db

(X) Which of the following combinations of logic gates can decode binary 1101?
a) One 4-input AND gate
b) One 4-input AND gate, one inverter
c) One 4-input AND gate, one OR gate
d) One 4-input NAND gate, one inverter

(XI) What is a transceiver circuit?
a) A buffer that transfers data from input to output.
b) A buffer that transfers data from output to input.

c) A buffer that can operate in both directions.
d) A buffer that can operate in one direction.

(XII) A binary-weighted digital-to-analog converter has an input resistor of 100 komega.gif. If the resistor is connected to a 5 V source, the current through the resistor is:
a) 50 micro ampere
b) 5 micro ampere
c) 500 micro ampere
d) 25 micro ampere

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Define conversion time and write down the disadvantages of digital ramp type analog to digital converter. [5]
3. Explain the operation of class A power amplifier and overall efficiency. [5]
4. Implement the following function with a 8:1 multiplexer [5]
   $F(A,B,C,D) = \Sigma (0,1,3,4,8,9,15)$
5. What is the function of shift register and write down the types of the shift registers based on applying inputs and accessing of outputs. [5]
6. An 8-bit DAC has an output of 3.92 mA for an input of 01100010. What are the DAC's resolution and full-scale output? What is the percentage resolution of an 8-bit DAC? [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. Design a 4 Bit synchronous binary Up-counter using J-K flip-flops and other necessary logic gates. [15]
8. a) What are the advantages of negative feedback ? [15]
   b) Explain the operation of a phase shift oscillator with circuit diagram.
   c) Derive an expression for its frequency of oscillation.
9. Explain the working principle of astable multivibrator using IC-555 and find out the frequency. [15]
10. Draw and explain a 16:1 Multiplexer using 4:1 Multiplexer [15]
11. Explain the working principle of Wien bridge oscillator and find out the frequency of oscillation. [15]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : HM-HU301/HSMC 301/HSMC-301/HSMC301 Economics for Engineers (Humanities-II)

UPID : 003446

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) The indirect method of preparing a statement of cash flows is also known as---------

(ii) Name the elements that may be included in the sunk costs.

(iii) The economic cost is not related with-----

(iv) The additional cost incurred for the production of one more unit of a product or service is called _____

(v) What is Opportunity Cost?

(vi) Rs. 1200 is lent out at 5% per annum simple interest for 3 years. What will be the amount after 3 years?

(vii) What is the Margin of Safety if sales is 20,000 units and B.E.P is 15,000 units?

(viii) Determine the stock turnover ratio if the opening stock is Rs 31,000, closing stock is Rs 29,000, sales is Rs 3,20,000 and gross profit ratio is 25% on sales.

a. 31 times
b. 11 times
c. 8 times
d. 32 times

(ix) The concept that the value of a rupee to be received in future is less than the value of a rupee on hand today is known as:

(x) Given that the selling price is Rs 20 per unit and variable cost is Rs 16 per unit, what will be the contribution of this unit?

(xi) Which index in India is used to evaluate inflation?

(xii) A company produces a single, high-volume product. During a given year its production volume was 780,000 units, fixed costs were Rs 32 Lac and variable costs were Rs. 16 per unit. What was the company's average fixed cost for the year?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. An electricity company wants to replace its machinery which was installed in the year 1982 at a cost of Rs 30,00,000 with a capacity of 300 MW. This cost consists of material, labour and overhead costs in the ratio of 5:3:2. The present cost index of materials, labour and overhead are 250, 300 and 240 respectively. The company wants to replace the existing machine with a machine of double capacity. Calculate the present cost of machinery to be replaced with double capacity by using cost indexes and power sizing model. The power sizing factor is given as 0.80. [5]

3. A Plant Manager is contemplating replacement policy to new equipment. The estimated costs are : [5]

Year	1	2	3	4	5	6
Replacement cost at the beginning of the year	100	120	130	150	160	200
Salvage value at the end of the year	50	40	35	20	10	0
Operating costs	30	40	50	65	80	90

Determine the year when replacement is to be made.

4. Write a note on cost indexes. [5]

5. From the following Balance sheets of Bengal Engineering Ltd, prepare a comparative balance sheet as on 31st March 2009 and 31st March 2010.

Liabilities	31/03/2009	31/03/2010	Assets	31/03/2009	31/03/2010
Share capital	800000	1000000	Fixed assets	2000000	2400000
Reserve and surplus	100000	200000	Less: provision for depreciation	400000	500000
12% debenture	1000000	1200000		1600000	1900000
Current liabilities	500000	700000	Current assets	800000	1200000
	2400000	3100000		2400000	3100000

6. A company purchased a Machine on 1st April 2012 for Rs. 2,60,000. The shipping and forwarding charges are Rs. 10,000. Import duty is Rs. 12,000 and installation charges are Rs. 10,000. Depreciation to be charged under diminishing balance method are as follows: [5]

1st year - 10% pa
2nd year - 20% pa
3rd year - 30% pa
Calculate the amount of depreciation and book value each year.

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. The following is the trial balance of Mr. R.P. Sharma for the year ended on 31st December 1998. Compile the Final accounts for him after attending to the adjustments. [15]

Heads of account	Debit (Rs)	Credit (Rs)
Land	100000	
Freehold premises	60000	
Bills receivable and payable	20000	24000
Wages	15000	
Rejections of goods including price adjustment	2500	3500
Personal accounts resulting from credit sales and purchases	31000	15000
Purchases and sales (including cash transactions)	40500	96000
Salaries	7000	
Insurance and municipal taxes	1500	
Rentals	2000	4500
Bad debts	600	
Carriage	3000	
Opening stock	5000	
Commission accounts	500	600
Vehicles	62000	
Machinery	85000	
Drawings and capital accounts	8000	175000
Loan from SFC		125000
Total	**443600**	**443600**

Adjustments:

a) Two thirds of the expenditure on carriage pertains to the goods purchased locally and imported.
b) Rs. 1000 due from Mr ghat is to be written off being irrecoverable .
c) Wage bill of Rs. 5,000 paid for an extension to the premises is accounted as wages paid and thus needs to be rectified.
d) Depreciate premises by 5%, vehicles by 20% and machinery by 5% .
e) Closing stock value is Rs 7500
f) Expenditure on salaries and rentals is payable to the extent of Rs 1000 and Rs 500 respectively. Insurance and municipal taxes are prepaid by Rs 200.

8. If the selling price per unit Rs 30, marginal cost per unit Rs. 15, fixed cost is Rs. 1,50,000, find out a) P.V ratio b) BEP c) BEP (in units) and d) Desired sales to earn a profit of Rs 150000. [15]

9. From the following particulars, prepare a statement in such form as you consider most suitable for showing clearly all elements of cost. [15]

	(Rs)
Opening stock of Raw Materials	25000
Purchase of raw materials	70000
Raw materials returned to suppliers	2000
Closing stock of Raw materials	18800
Wages paid to Productive workers	18000
Wages paid to Non productive workers	2000
Salaries paid to office staff	5000
Carriage on Raw materials purchased	500
Carriage on goods sold	1500
Rent and rates of workshop	2500
Fuel,gas water etc	1000
Repairs to plant	600
Depreciation on Machinery	1400
Office expenses	1500
Direct chargeable expenses	800
Advertising	1200
Abnormal loss of materials	1200

10. A company has to make a choice between two projects namely A and B. The initial capital outlay of the two projects are Rs. 1,35,000 and Rs. 24,0000 respectively for A and B. There will be no scrap value at the end of the life of either of the projects. The opportunity cost of capital of the company is 16%. The annual incomes in Rs. are as follows: [15]

Year	Project A	Project B
1	NIL	60,000
2	30,000	84,000
3	1,32,000	96,000
4	84,000	1,02,000
5	84,000	90,000

Calculate discounted payback period and profitability index .

11. The trial balance of a business as on 31/12/1998 is given below : [15]

Heads of Account	Debit (Rs)	Credit(Rs)
Returns	25,900	10,900
Discount received		5200
Sales		6,12,700
Commission		11,500
Bills receivable	65,300	
Carriage inwards	14,900	
Plant and Machinery	85,600	
Investment	22,400	
Sundry Creditors		60,000
Salaries	15,500	
Freehold premises	1,05,000	
Capital		1,50,000
Purchase	3,40,000	
Reserve		75,000
Wages	65,800	
Discount allowed	6,500	
Sundry debtors	85,800	
Freight	12,400	
Bills payable		45,800
Furniture	48,600	

Carriage outwards	15,200	
Stock(opening)	35,900	
Cash at Bank	24,700	
Cash in hand	1,600	
Total	**9,71,100**	**9,71,100**

Prepare Trading and Profit and loss account for the year ended 31/12/1988 and a balance sheet as on that date, after taking into consideration the following adjustments:

a) Closing stock Rs. 85,000

b) Outstanding liabilities for salaries Rs. 4,500 and wages Rs. 9,800

c) Depreciate furniture by 15% and plant and machinery by 20%.

d) Sundry debtors include Rs. 1,800 as bad which is to be written off.

e) Create a provision for doubtful debts @ 5% on sundry debtors.

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PC-ECS302/PCC- CS302/PCC-CS302/PCCCS302 Computer Organisation

UPID : 003444

Time Allotted : 3 Hours *Full Marks : 70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]
   (i) In case of Zero-address instruction method the operands are stored in _____.
   (ii) How many memory locations can be addressed by a 32-bit computer?
   (iii) What is page fault?
   (iv) What is Microprogrammed control unit ?
   (v) The addressing mode of an instruction is resolved by _____.
   (vi) What is I/O interface ?
   (vii) What is Hit ratio in cache memory?
   (viii) What is interrupt-service-routine?
   (ix) CPU consists of_____
   (x) Overflow occurs when_____
   (xi) Associative memory is a_____
   (xii) What is INTR?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. What is Carry Look Ahead Adder? [5]
3. What is virtual memory ? Write the advantages of virtual memory. [5]
4. What are 'write through' and 'write back' policies in cache memory? [5]
5. Differentiate isolated I/O and memory mapped I/O. [5]
6. Multiply +12 and +14 using Booth's algorithm. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Compare RISC with CISC. [7]
   (b) Divide 9 by 2 using restoring division method. [8]
8. Describe various addressing modes with example. [15]
9. (a) Explain DMA method with block diagram . [7]
   (b) Explain Bus Arbitration techniques. [8]
10. (a) Evaluate the arithmetic statement X = (A*B)/(C+D) in one, two and three address machines. [9]
    (b) Explain Instruction Cycle with suitable flow chart. [6]
11. (a) What are the different types of interrupt? Give examples. [8]
    (b) What are the differences between vectored and non-vectored interrupt? [7]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PC-ECS301/PCC- CS301/PCC- CSBS301/PCC-CS301/PCCCS301 Data Structure & Algorithms

UPID : 003443

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) What is the maximum number of swaps that can be performed in the Selection Sort algorithm?
a) n - 1
b) n
c) 1
d) n - 2

(ii) In infix to postfix conversion algorithm, the operators are associated from?
a) right to left
b) left to right
c) centre to left
d) centre to right

(iii) What is the disadvantage of array data structure?
a) The amount of memory to be allocated should be known beforehand.
b) Elements of an array can be accessed in constant time.
c) Elements are stored in contiguous memory blocks.
d) Multiple other data structures can be implemented using arrays.

(iv) Which of the following data structures finds its use in recursion?
a) Stack
b) Arrays
c) Linked List
d) Queues

(v) Which of the following sorting algorithms provide the best time complexity in the worst-case scenario?
a) Merge Sort
b) Quick Sort
c) Bubble Sort
d) Selection Sort

(vi) From the following which is not the operation of data structure?
a) Operations that manipulate data in some way.
b) Operations that perform a computation.
c) Operations that check for syntax errors.
d) Operations that monitor an object for the occurrence of a controlling event.

(vii) What is the information, which a LinkedList's Node must store?
a) The address of the next node if it exists
b) The value of the current node
c) Both (A) and (B)
d) None of the above

(viii) Which of the following represents the Postorder Traversal of a Binary Tree?
a) Left -> Right -> Root
b) Left -> Root -> Right
c) Right -> Left -> Root
d) Right -> Root -> Left

(ix) In a graph of 'n' nodes and 'n' edges, how many cycles will be present?
a) Exactly 1
b) At most 1
c) At most 2
d) Depends on the graph

(x) Which of the following represents the Postorder Traversal of a Binary Tree?
a) Left -> Right -> Root
b) Left -> Root -> Right
c) Right -> Left -> Root
d) Right -> Root -> Left

(xi) Which of the following is the most widely used external memory data structure?
a) AVL tree
b) B-Tree
c) Red-black Tree
d) Both AVL Tree and Red-black Tree

(xii) Which one of the following data structures are preferred in database-system implementation?
a) AVL Tree
b) B Tree
c) B+ Tree
d) Splay Tree

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Convert the given infix expression to postfix expression [5]
A+B*C-D$F/H

3. Write an algorithm to convert infix expression to postfix expression. [5]

4. Is implementation of queue by circular linked list is better than by linear linked list? Justify your answer. [5]

5. Justify the statement with proper example: All Binary Search Trees are Binary Tree but All Binary Trees are not Binary Search Tree. [5]

6. Write an algorithm to insert an element in Circular Queue. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. a) Define Circular Linked List and Double Linked List. [3+4+4+4]
b) Write an algorithm to delete an element from Double Linked List.
[Consider three cases: 1st node deletion, last node deletion and any intermediate node deletion]

8. a)Describe Binary Search algorithm with a suitable example. [6+4+5]
b)Explain with example of Insertion and Deletion operation in Threaded Binary Tree.
c)Explain Double Hashing.

9. a)Describe heap sort algorithm with a suitable example. [7+8]
b)Compare the Complexity Analysis [Best Case, Avg. Case, Worst Case] of the following sorting algorithms; Bubble Sort, Selection, Insertion Sort, Quick Sort, Merge Sort, Heap Sort, Radix Sort

10. a)Which algorithm is better in between iterative and recursive algorithms?----- Justify your statement. [4+6+5]
b)Write an algorithm to perform the following operation in Circular Linked List; Insert an element in first position in the linked list; Delete the last element of the linked list
c) Explain the insertion mechanism in B+ Tree.

11. Write short notes on any three from following: [15]
a) AVL Tree
b) Dequeue and Priority Queue
c) Collison Resolution Techniques in Hashing
d) BFS

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-M301/BSC 301/BSC301 Mathematics-III (Differential Calculus)

UPID : 003445

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is the nature of the sequence $\{(-1)^n\}$

(II) If $z = \log(x^3 + y^3)$ then find the value of $\frac{\partial z}{\partial y}$ at (1,1)

(III) Evaluate $\int_0^{\frac{\pi}{2}}\int_0^{\pi} \sin(x+y)dxdy$

(IV) What is the order and degree of the ODE $(\frac{d^2y}{dx^2} + \frac{dy}{dx})^{\frac{3}{2}} = 5$

(V) Find C.F for the equation $(D^2 - 5D + 6)y = 0$

(VI) Define pseudo graph with an example.

(VII) Check the convergence of $\sum_{n=1}^{\infty} \frac{2}{e^n}$

(VIII) If $x = r\cos\theta$, $y = r\sin\theta$ then what is the value of $\frac{\partial(r,\theta)}{\partial(x,y)}$

(IX) Evaluate $\int_0^{\frac{\pi}{2}}\int_0^{\pi} \cos(x+y)dxdy$

(X) The ODE $(ax^2 + by + c)dx + (cy^2 + dx + e)dy = 0$ is exact then write the relation in between a, b and c

(XI) Find the values of x such that the sequence $\{nx^n\}$ is null sequence.

(XII) For what values of p, the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ will be divergent.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Prove that the maximum number of edges in a connected simple graph with n number of vertices is n(n-1)/2. [5]

3. Prove that the sequence [5]

$\{\sqrt{7}, \sqrt{7+\sqrt{7}}, \sqrt{7+\sqrt{7+\sqrt{7}}}, \ldots\ldots\infty\}$

converges to the positive root of

$x^2 - x - 7 = 0.$

4. Show that $\vec{A} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational. Find the scalar function ϕ such that $\vec{A} = \vec{\nabla}\phi$

5. Verify Green's theorem in the plane for $\oint_C (xy + y^2)dx + x^2 dy$ where C is the closed curve of the region bounded by y=x and $y=x^2$ [5]

6. Solve: $p^2 + 2py\cot x = y^2$.

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) State Gauss and Stokes theorem of vector calculus. [5]

(b) Evaluate: [5]

Evaluate the line integral $\int (x^2 + xy)\,dx + (x^2 + y^2)\,dy$, where c is the square formed by the lines x=1,x=-1,y=1,y=-1.

(c) Evaluate: [5]

$\iiint (x + y + z + 1)^4 \, dxdydz$ over the region defined by $x \geq 0, y \geq 0, z \geq 0, x + y + z \leq 1$

8. (a) Answer the following: [5]

Solve $2ydx - xdy = xy^3 dy$

(b) Answer the following: [5]

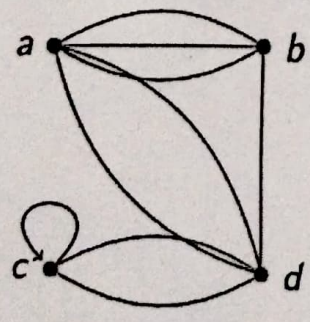
Define integrating factor (I.F) and solve $(x^2y - 2xy^2)dx + (3x^2y - x^3)dy = 0$

(c) Answer the following: [5]

Solve $x\frac{dy}{dx} + y = y^2 \log x$

9. (a) Answer the following: [8]

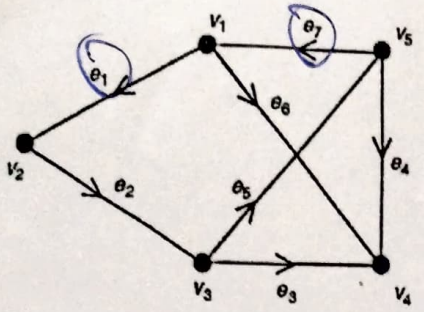
Find adjacency matrix from the graph



(b) Answer the following: [7]

nction

rite the incidence matrix from the following Di-graph



10. (a) Find the directional derivative of $f=xy^2 + yz^3$ at the point (2, -1, 1) in the direction of the vector $i+2j+2k$. [5]

(b) Check whether the vector $A= 3y^4z^2\,i + 4x^3z^2\,j - 3x^2y^2\,k$ is solenoidal or not. [5]

(c) Answer the following: [5]

Find P.I $(D^2-1)y=\sin^2 x$

11. (a) Find the interval and radius of convergence of the power series [5]

$$\sum_{n=1}^{\infty}(-1)^n\frac{10^n}{n!}(x-10)^n$$

(b) Answer the following: [5]

Expand $f(x)=\sin x$ in an infinite series in power of x

(c) Answer the following: [5]

Expand $f(x)=e^x$ in an infinite series in power of x

*** END OF PAPER ***